

Week 10: Learning – with Latent Random Variables

Practical Assignment

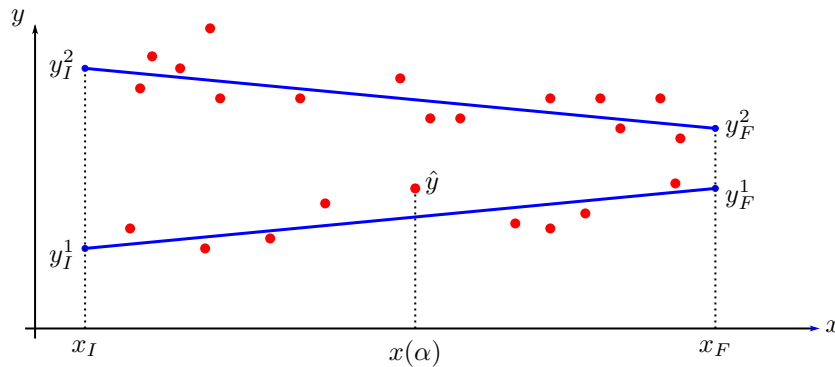
For Weeks 8 and 9, we looked at how to estimate the parameters of a Bayes network (BN) from data, where we assumed that the data consists of observations for *all* the random variables (RVs) in the model. For this week, we relax this assumption and we look at how to estimate parameters when some of the RVs in the model are *latent* (i.e., *unobserved* or *hidden*). The technique we use to do this is the *expectation-maximisation (EM) algorithm*.

The practical assignment for this week applies the EM algorithm to the following two problems:

- An extension of the line-fitting problem of Weeks 8 and 9 – we now fit *multiple* lines to the data
- Revisiting the estimation of parameters for the Hamming (7,4) error-correction code of Week 8, but with the transmitted bits hidden (latent/unobserved) – this is an optional question that you can complete for extra credit

1 EM for a regression example with multiple lines

For this question, we put a twist on the line-fitting problem of the past few assignments. In the previous assignments, the data points were noisy measurements of a single line, and the y -coordinates of the endpoints of the line were unknown – these coordinates were the parameters to be estimated. In this assignment, we extend the problem to one where a data point is a noisy measurement of one of L different lines, and we try to estimate the y -coordinates of all the lines from the measurements. Consider the figure below with $L = 2$ lines, and with the y -coordinates of both lines shown.



We know that there are two lines, but we do not know which of the two lines generated any specific measurement. We model this uncertain association between the lines and a measurement with a discrete RV z , where the value of z refers to the line that generated the measurement. For this two-line case, the measurement model of the previous assignments is now extended to become

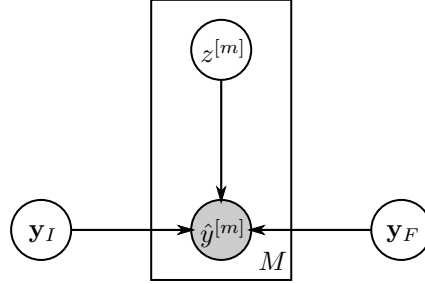
$$\hat{y} = \begin{cases} (1 - \alpha)y_I^1 + \alpha y_F^1 + v, & \text{if } z = 1 \\ (1 - \alpha)y_I^2 + \alpha y_F^2 + v, & \text{if } z = 2 \end{cases}, \quad (1)$$

where the measurement noise for any point is given by $v \sim \mathcal{N}(v; 0, \sigma_v^2)$. A measurement is therefore drawn from a different Gaussian distribution when line 1 generates the measurement than when line 2 generates the measurement.

In the general case where there are L lines, the y -coordinates of the left endpoints of the lines are given by $\mathbf{y}_I = (y_I^1, \dots, y_I^L)$ and the y -coordinates of the right endpoints are given by $\mathbf{y}_F = (y_F^1, \dots, y_F^L)$. There are M

association variables – one for each measurement – and the m th association variable is a discrete RV that can take on L possible values, or $z^{[m]} \in \{1, \dots, L\}$. The quantities of the general description of the EM algorithm of Section 3 in this week's notes are the following: the parameters are the endpoints of the lines, $\Theta = (\mathbf{y}_I, \mathbf{y}_F)$, the hidden variables are the M association variables, $\mathcal{H} = (z^{[1]}, \dots, z^{[M]})$, and the data consists of the noisy measurements, $\mathcal{D} = (\hat{y}^{[1]}, \dots, \hat{y}^{[M]})$. The objective of this question is to estimate the parameters Θ from the data $\mathcal{D}$ using the EM algorithm.

We can model this setup using a BN in plate notation as shown below.



From the BN, we can see that each measurement $\hat{y}^{[m]}$ has an association variable $z^{[m]}$ that functions as a “selector” of the lines parameterised by $\mathbf{y}_I$ and $\mathbf{y}_F$. We assume that each line is equally likely to be selected, which means that $p(z^{[m]}) = \frac{1}{L}$ (i.e., the prior distribution over $z^{[m]}$ is uninformative or flat).

Since working out the EM algorithm for this example is somewhat involved, we will work through most of the development together; you will only need to complete the last bit of the development and implement the algorithm in code by yourself.

E-step: For the E-step, we have to calculate the posterior distribution over the hidden variables $\mathcal{H}$ given the data $\mathcal{D}$ and a specific set of parameters Θ ; i.e., we should calculate $p(\mathcal{H}|\mathcal{D}, \Theta)$. This means that we have to perform inference for this model.

Since the prior distributions over $z^{[m]}$ are uninformative, they have no effect on inference, and we can omit them. The only factors of interest are therefore the likelihood functions for the measurements; the likelihood function of the m th measurement is given in the table below.

$z^{[m]}$	$p(\hat{y}^{[m]} \mathbf{y}_I, \mathbf{y}_F, z^{[m]})$
1	$\mathcal{N}(\hat{y}^{[m]}; (1 - \alpha^{[m]})y_I^1 + \alpha^{[m]}y_F^1, \sigma_v^2)$
$\vdots$	$\vdots$
L	$\mathcal{N}(\hat{y}^{[m]}; (1 - \alpha^{[m]})y_I^L + \alpha^{[m]}y_F^L, \sigma_v^2)$

As you can see, there is a different Gaussian likelihood function for each value of the association variable $z^{[m]}$. The combined likelihood function is the product of the likelihood functions for all the measurements, or

$$p(\mathcal{D}|\mathcal{H}, \Theta) = \prod_{m=1}^M p(\hat{y}^{[m]}|\mathbf{y}_I, \mathbf{y}_F, z^{[m]}). \quad (2)$$

The posterior distribution over the hidden variables is now simply given by

$$p(\mathcal{H}|\mathcal{D}, \Theta) \propto p(\mathcal{D}|\mathcal{H}, \Theta) p(\mathcal{H}), \quad (3)$$

and since $p(\mathcal{H})$ is uninformative, the posterior is directly proportional to the combined likelihood. The only remaining step in this inference process is to normalise the result. Since the association variables are independent of each other in the posterior distribution (you can clearly see this from the BN, or from the factorisation of Equation 3 as given by Equation 2), we can normalise the factor over each association variable $z^{[m]}$ separately, which results in the following table

$z^{[m]}$	$q(z^{[m]}) = p(z^{[m]} \hat{y}^{[m]}, \mathbf{y}_I, \mathbf{y}_F)$
1	$\gamma_1^{[m]}$
$\vdots$	$\vdots$
L	$\gamma_L^{[m]}$

where

$$\gamma_l^{[m]} = \frac{\mathcal{N}(\hat{y}^{[m]}; (1 - \alpha^{[m]}) y_I^l + \alpha^{[m]} y_F^l, \sigma_v^2)}{\sum_{i=1}^L \mathcal{N}(\hat{y}^{[m]}; (1 - \alpha^{[m]}) y_I^i + \alpha^{[m]} y_F^i, \sigma_v^2)}. \quad (4)$$

The values $\gamma_1^{[m]}, \dots, \gamma_L^{[m]}$ are called the *responsibilities*, and they represent the soft association between data point m and each of the lines. Note that there are $M \times L$ responsibilities in total.

As specified in the E-step, we set $q(\mathcal{H})$ equal to this posterior; i.e.,

$$q(\mathcal{H}) = \prod_{m=1}^M q(z^{[m]}) = \prod_{m=1}^M p(z^{[m]} | \hat{y}^{[m]}, \mathbf{y}_I, \mathbf{y}_F), \quad (5)$$

where $p(z^{[m]} | \hat{y}^{[m]}, \mathbf{y}_I, \mathbf{y}_F, \cdot)$ is specified in the table above. The distribution $q(\mathcal{H})$ is therefore parameterised by the $M \times L$ responsibilities.

M-step: The aim of the M-step is to construct the auxiliary function $\mathcal{Q}(\Theta)$ using distribution $q(\mathcal{H})$, and then to find the parameters Θ that maximise this function.

We start this step by writing down the joint distribution $p(\mathcal{D}, \mathcal{H} | \Theta)$, which is simply the product of factors in the model,

$$p(\mathcal{D}, \mathcal{H} | \Theta) = \prod_{m=1}^M p(\hat{y}^{[m]} | \mathbf{y}_I, \mathbf{y}_F, z^{[m]}) p(z^{[m]}). \quad (6)$$

The auxiliary function $\mathcal{Q}(\Theta)$ is then given by

$$\begin{aligned} \mathcal{Q}(\Theta) &= \sum_{\mathcal{H}} q(\mathcal{H}) \log p(\mathcal{D}, \mathcal{H} | \Theta) \\ &= \sum_{z^{[1]}, \dots, z^{[M]}} \left[\prod_{i=1}^M q(z^{[i]}) \right] \left[\sum_{m=1}^M \log p(\hat{y}^{[m]} | \mathbf{y}_I, \mathbf{y}_F, z^{[m]}) + \sum_{m=1}^M \log p(z^{[m]}) \right] \\ &= \sum_{m=1}^M \left\{ \sum_{z^{[1]}, \dots, z^{[M]}} \left[\prod_{i=1}^M q(z^{[i]}) \right] \log p(\hat{y}^{[m]} | \mathbf{y}_I, \mathbf{y}_F, z^{[m]}) \right\} + \sum_{z^{[1]}, \dots, z^{[M]}} \left[\prod_{i=1}^M q(z^{[i]}) \right] \left[M \log \frac{1}{L} \right], \end{aligned} \quad (7)$$

where we used Equations 5 and 6, moved the inner sum to the outside, and used the fact that the prior distribution $p(z^{[m]})$ is uninformative. We can manipulate the last term by moving each of the sums one by one into the product, or

$$\sum_{z^{[1]}, \dots, z^{[M]}} \left[\prod_{i=1}^M q(z^{[i]}) \right] = \left[\sum_{z^{[1]}} q(z^{[1]}) \right] \times \dots \times \left[\sum_{z^{[M]}} q(z^{[M]}) \right] = 1, \quad (8)$$

which reduces to 1 since the marginalisation of distribution $q(z^{[m]})$ over its variable $z^{[m]}$ is 1. We now continue manipulating $\mathcal{Q}$ by moving each of the sums one by one into the product, and using the same marginalisation property on all but one of the factors:

$$\begin{aligned} \mathcal{Q}(\Theta) &= \sum_{m=1}^M \left\{ \left[\sum_{z^{[1]}} q(z^{[1]}) \right] \times \dots \times \left[\sum_{z^{[m]}} q(z^{[m]}) \log p(\hat{y}^{[m]} | \mathbf{y}_I, \mathbf{y}_F, z^{[m]}) \right] \times \dots \times \left[\sum_{z^{[M]}} q(z^{[M]}) \right] \right\} \\ &\quad + \text{const.} \\ &= \sum_{m=1}^M \left\{ \sum_{z^{[m]}} q(z^{[m]}) \log p(\hat{y}^{[m]} | \mathbf{y}_I, \mathbf{y}_F, z^{[m]}) \right\} + \text{const.} \end{aligned} \quad (9)$$

The value of the constant terms are unimportant, since it will not effect the subsequent optimisation. We can now swap the sums around, and write things in terms of the responsibilities $\gamma_l^{[m]}$ and the definition of the likelihood

function for each measurement (as defined in the development in the E-step) as follows:

$$\begin{aligned} \mathcal{Q}(\Theta) &= \sum_{z^{[m]}} \left\{ \sum_{m=1}^M q(z^{[m]}) \log p(\hat{y}^{[m]} | \mathbf{y}_I, \mathbf{y}_F, z^{[m]}) \right\} + \text{const.} \\ &= \sum_{l=1}^L \left\{ \sum_{m=1}^M \gamma_l^{[m]} \left(-\frac{1}{2\sigma_v^2} \left[\hat{y}^{[m]} - (1 - \alpha^{[m]}) y_I^l - \alpha^{[m]} y_F^l \right]^2 \right) \right\} + \text{const.} \end{aligned} \quad (10)$$

If you look closely at Equation 10, you will see that each term only depends on the parameters of a single line (or is constant). The part between braces ($\{\cdot\}$) should also look familiar to you – it is virtually the same as the log-likelihood function used for ML estimation for a single line – only the responsibilities are new.

At this point, our journey together comes to an end and I leave you to forge ahead by yourself – you have to complete the M-step by yourself.

Note that for the E-step in the EM algorithm, it is necessary to perform inference for *every* data point. However, the inference for this problem is trivial (it amounts to the calculation of a responsibility for each line in each data point as given in Equation 4), and we do not need `emdw`'s functionality for this question. You therefore do not have to use C++ (I suggest you use the same software setup as for Question 2 of Week 8 – you should be able to reuse much of your code here).

- (a) (On paper) Complete the development of the M-step; i.e., calculate the parameters Θ^{new} that would maximise $\mathcal{Q}(\Theta)$, or

$$\Theta^{\text{new}} = \arg \max_{\Theta} \mathcal{Q}(\Theta). \quad (11)$$

Note that you can do the optimisation for each line separately, since each term in $\mathcal{Q}(\Theta)$ only depends on the parameters of a single line (y_I^l and y_F^l). Use a similar optimisation procedure for each line than what you previously used for MLE for a single line (Question 2 of Week 8).

- (b) (In code) Choose the number of lines L as well as the parameters for each line ($\mathbf{y}_I$ and $\mathbf{y}_F$). Now simulate a number of measurements by first drawing a value for $z^{[m]}$ and then drawing a value for $\hat{y}^{[m]}$ given the value of $z^{[m]}$ and the parameters of the associated line. Plot the actual lines and measurements.
- (c) (In code) Implement the EM algorithm: Choose appropriate initial parameter values, and then iteratively perform the E-step and M-step until the parameters have converged. Plot the estimated lines, the actual lines and the data points on the same graph for some of the iterations.
- (d) (In code) Investigate the effect of parameter initialisation, measurement noise level, number of measurements, as well as the number and arrangement of lines on the performance of the EM algorithm.

2 EM for the Hamming (7,4) error-correction code (optional)

For this optional question (that you can complete for bonus marks), we'll revisit the familiar Hamming (7,4) error-correction code. In Question 1 of Week 8, you used MLE to estimate the parameters of the check bit and measurement factors with the assumption that both the sent bits $b_0, \dots, b_6$ and the received bits $r_0, \dots, r_6$ are known. Now repeat this question, but with only the received bits observed; the sent bits are therefore latent/hidden RVs. Since we have latent RVs, you will have to use the EM algorithm for estimation.

Remember that in the E-step of the EM algorithm, you have to do inference for *every* data point. Since inference for the Hamming (7,4) error-correction code is not trivial, you have to use `emdw` for this question.

- (a) (On paper) Draw the BN for the model with unknown parameters and for M data points using plate notation. Decompose the objective of the E-step, $q(\mathcal{H}) = p(\mathcal{H} | \mathcal{D}, \Theta^{\text{old}})$, in terms of the latent variables for each data point, $b_0^{[m]}, \dots, b_6^{[m]}$, the observed variables for each data point, $r_0^{[m]}, \dots, r_6^{[m]}$, and parameters for this model. (Can you see why you need to do inference for every data point, and how you can use your work in previous practicals to do this?)
- (b) (On paper) Write $\log p(\mathcal{D}, \mathcal{H} | \Theta)$ as a sum of local log-likelihood functions. Now manipulate the auxiliary function of the M-step, $\mathcal{Q}(\Theta) = \sum_{\mathcal{H}} q(\mathcal{H}) \log p(\mathcal{D}, \mathcal{H} | \Theta)$, to be a sum of local log-likelihood functions combined with a marginal belief of $q(\mathcal{H})$. Further manipulate each term such that it is written in terms of

table entries in the marginal belief of $q(\mathcal{H})$ and the parameters of the local log-likelihood function. For example, the term corresponding to the local log-likelihood function $\log p(r_i | b_i : \theta_e)$ should be written only in terms of $q(b_i^{[m]} = r_i^{[m]})$, $q(b_i^{[m]} \neq r_i^{[m]})$ and θ_e (this should be somewhat similar to Question 1(c) of Week 8).

- (c) (On paper) Find the parameters that maximise the likelihood (in terms of the number of each type of observation) by solving for $\frac{\partial}{\partial \theta_k} \mathcal{Q}(\Theta) = 0$ for each parameter $\theta_k \in \Theta$.
- (d) (In code) Similar to Question 1 of Week 8, generate M data points, but consider the transmitted bits, $b_0, \dots, b_6$, to be hidden (i.e., after you have generated the data, knowledge of the transmitted bits may not be used in parameter estimation). Implement the EM algorithm to estimate Θ . You can first assume that the parameters of the check bit factors are known (i.e., only the parameter of the measurement factors is unknown), and then extend it to include estimation of the check bit parameters too.
- (e) (In code) Repeat the sampling of data points and parameter estimation, but vary the number of data points. Also vary the initialisation of the parameters. What are the effects of these variations in the number of samples and parameter initialisation of the accuracy of the parameter estimates and the behaviour of the EM algorithm?

3 Evaluation: Write and submit a short report

Document the following in a short report:

- Present an *outline* of the derivation for Question 1(a) [4 marks]
- Present an *overview* of your implementation for Question 1(c) [2 marks]
- Present your experiments, results and conclusions for Question 1(d) [4 marks]
- (Optional) Present your theoretical development, implementation, experiments, results and conclusions for Question 2 [10 bonus marks]

Submit both the report (in PDF format) and all your code for this assignment on STEMLearn by **Sunday 3 May at 23:59**.

See the DE424 info sheet for general guidelines for the practical report. Also see the document about how to avoid academic misconduct in DE424 on STEMLearn. Specific instructions:

- The text in the report should be typed, the page limit is **5 pages**¹, and the report must be submitted in PDF format.
- At the top of the report, you should include a declaration that the report and the work presented in it are your own work. If you received any help or information from any source that contributed to the work or report, you should also mention the source and the extent of the help or information in the declaration.
- The report should be clear, but you do not have to spend much time on making it beautiful. You can, for instance, draw diagrams or write mathematical manipulations by hand, scan it in, and include it in your report as an image (instead of drawing diagrams in applications such as Inkscape or typing the mathematical manipulations as equations in $\text{\LaTeX}$).
- You should give enough context in your report so that someone who is familiar with the module can understand how you have solved the problem and how well it worked without having to look at your code. Make sure that all your design decisions are clearly documented (and – if necessary – motivated). However, you should also be concise: avoid giving lengthy discussions of obvious matters, showing minute steps in mathematical manipulations, or copying large parts of your code into your report.

¹This page limit is applicable to the compulsory part of the report. If you choose to do the optional question, then *the part of your report that presents your work on the optional question* may exceed the page limit.